!

Strong and Weak Acids/Bases

The strength of an acid (or base) is determined by the amount of IONIZATION.

$$HCl(aq) + H_2O(\ell) \longrightarrow H_3O^+(aq) + Cl^-(aq)$$

hydrochloric acid
strong electrolyte
= 100% ionized

water

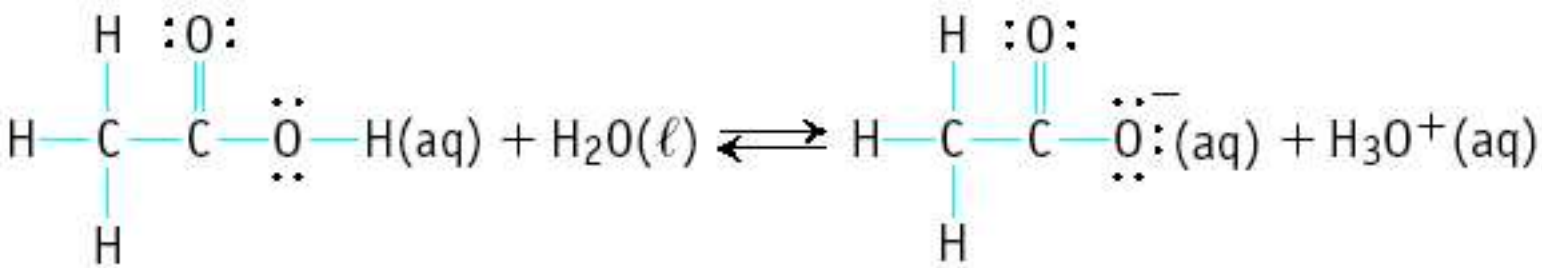
hydronium ion

chloride ion

HNO_3, HCl, H_2SO_4 and $HClO_4$ are among the only known strong acids.

- Weak acids are much less than 100% ionized in water.

One of the best known is acetic acid = CH_3CO_2H



- Strong Base: 100% dissociated in water.

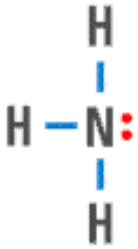
$NaOH\ (aq) \longrightarrow Na^{+}\ (aq) + OH^{-}\ (aq)$

Other common strong bases include KOH and $Ca(OH)_2$.

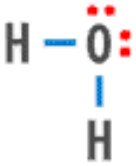
- Weak base: less than 100% ionized in water

One of the best known weak bases is ammonia

NH_3 (g) + H_2O (l) → NH_4^+ (aq) + OH^- (aq)



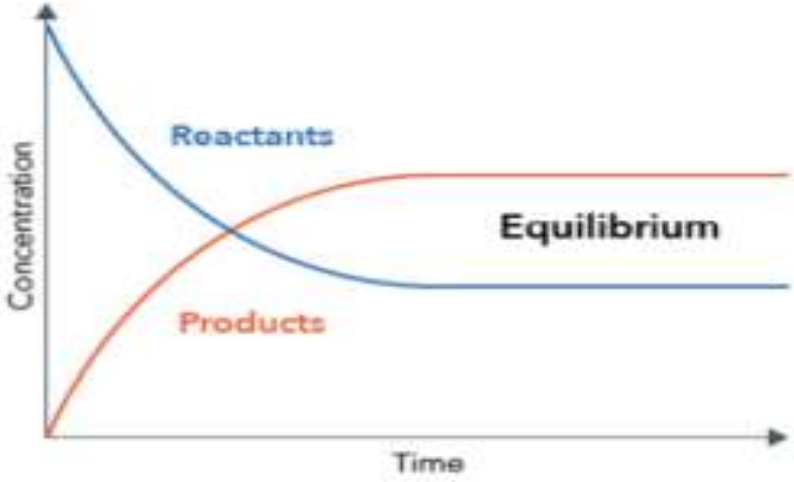
Base



Acid

Chemical Equilibrium

- Is a state in which the **rate** of the forward reaction equals the rate of the backward reaction
- The concentrations of reactants and products **do** not change at equilibrium.
- Only occur in **reversible reaction**



Example:

$$N_2O_4(g) \rightleftharpoons 2NO_2(g)$$

At equilibrium: $\text{rate}_{\textbf{forward}} = \text{rate}_{\textbf{reverse}}$

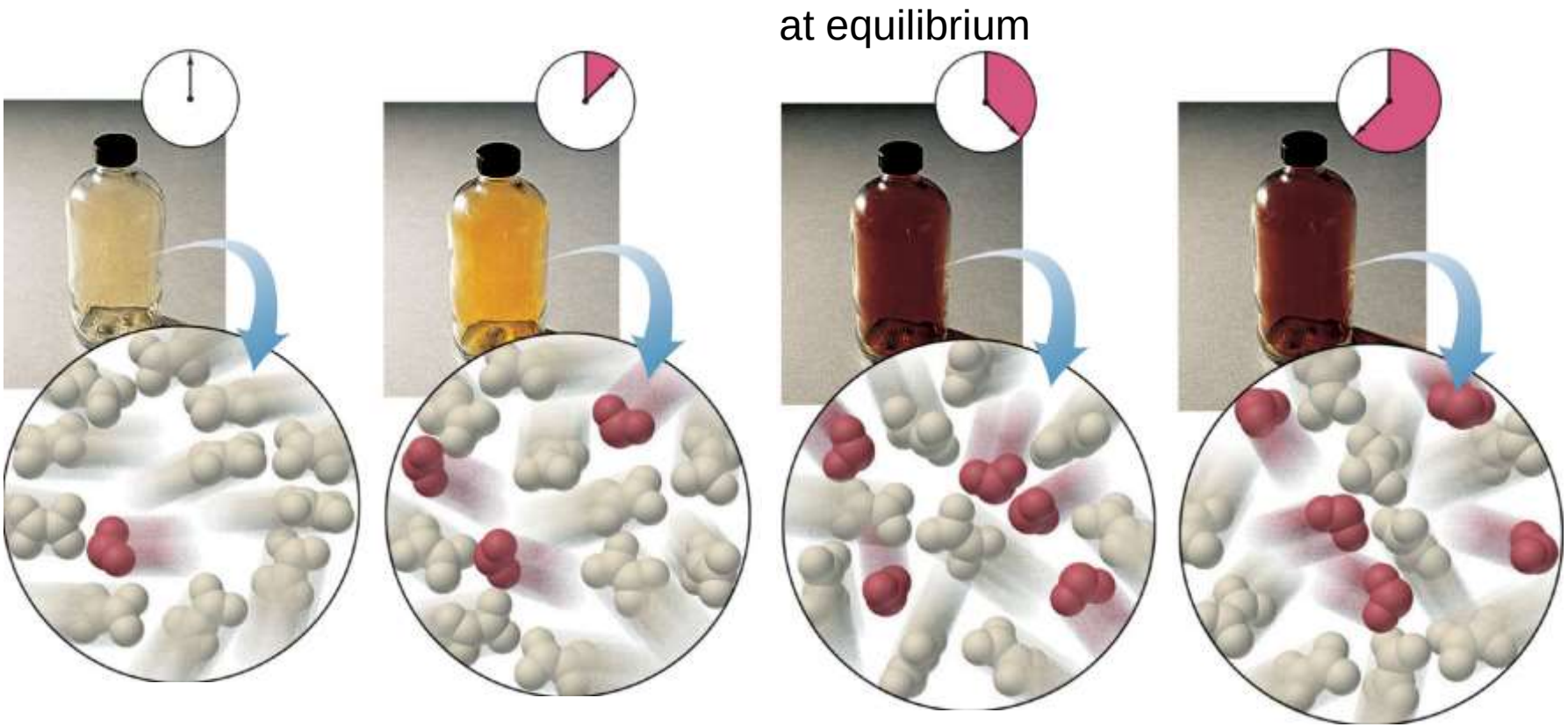
$$k_{\text{fwd}}[N_2O_4]_{\text{eq}} = k_{\text{rev}}[NO_2]^2_{\text{eq}}$$

$$k_{\text{fwd}}/k_{\text{rev}} = [NO_2]^2_{\text{eq}} / N_2O_4]_{\text{eq}} = K$$

K = the equilibrium constant

The magnitude of K indicates how far a reaction proceeds toward product at a given temperature.

Reaching Equilibrium on the Macroscopic and Molecular Levels



$$N_2O_4(g) \rightleftharpoons 2NO_2(g)$$

$N_2O_4(g)$ is colorless and $NO_2(g)$ is brown

Expression for Equilibrium Constant

Consider the following equilibrium system:

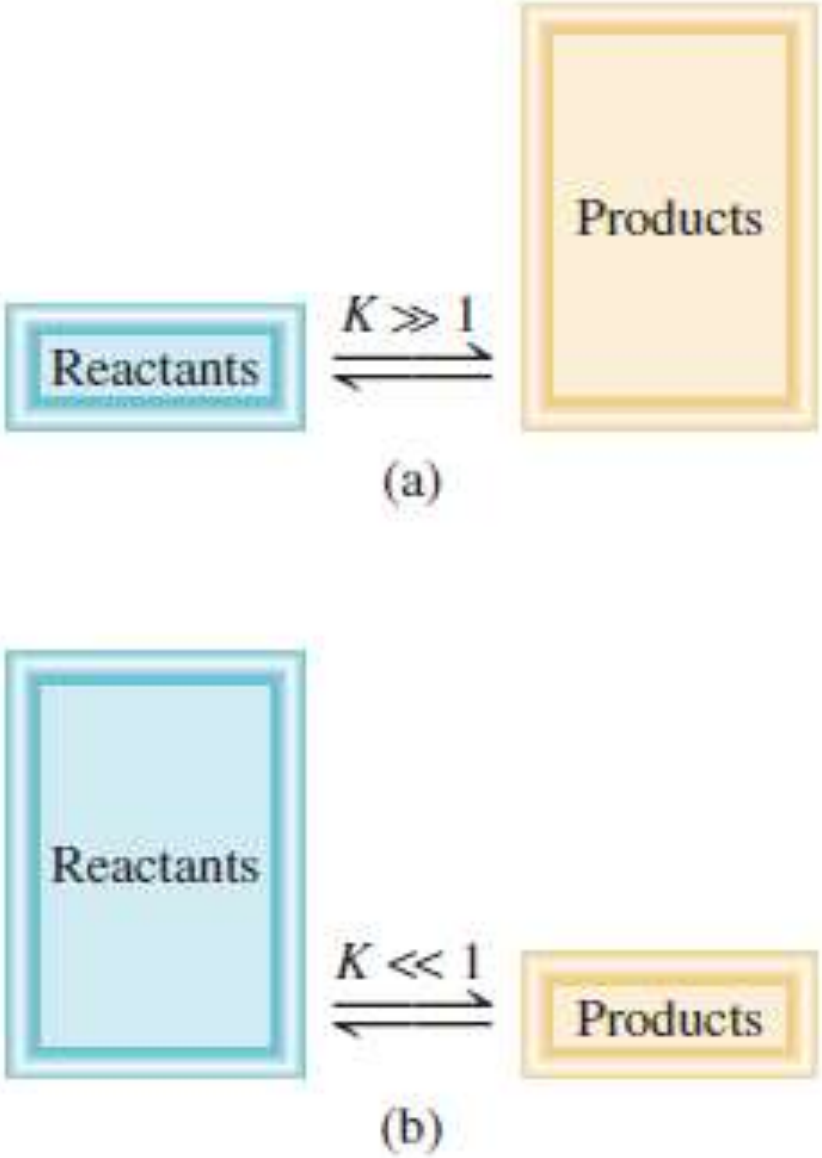
$$wA + xB \rightleftarrows yC + zD$$

$$K_c = \frac{[C]^y[D]^z}{[A]^w[B]^x}$$

- The numerical value of K_c is calculated using the concentrations of reactants and products that **exist at equilibrium**.

Chemical Equilibrium

- Chemical equilibrium is defined by K.
- The magnitude of K will tell us if the equilibrium reaction favors the reactants or the products.
  - a)If K » 1……..favors products
  - b)If K « 1……..favors reactants


Products
Reactants
$K \gg 1$
(a)
Reactants
$K \ll 1$
Products
(b)

Expressions for Equilibrium Constants

- Examples:

$$N_{2(g)} + 3H_{2(g)} \rightleftarrows 2NH_{3(g)}; \quad K_c = \frac{[NH_3]^2}{[N_2][H_2]^3}$$

$$PCl_{5(g)} \rightleftarrows PCl_{3(g)} + Cl_{2(g)}; \quad K_c = \frac{[PCl_3][Cl_2]}{[PCl_5]}$$

Determining equilibrium concentrations from K_c

PROBLEM: In a study concerning the conversion of methane to other fuels, a chemical engineer mixes gaseous CH_4 and H_2O in a 0.32 L flask at 1200 K. At equilibrium the flask contains 0.26 mol of CO, 0.091 mol of H_2, and 0.041 mol of CH_4. What is the $[H_2O]$ at equilibrium? K_c = 0.26 for the equation:

$$CH_4(g) + H_2O(g) \rightleftharpoons CO(g) + 3H_2(g)$$

PLAN: Use the balanced equation to write the K_c expression, and then substitute values for each component.

SOLUTION:

$$CH_4(g) + H_2O(g) \rightleftharpoons CO(g) + 3H_2(g) \qquad K_c = \frac{[CO][H_2]^3}{[CH_4][H_2O]}$$

$$[CH_4]_{eq} = \frac{0.041 \text{ mol}}{0.32 \text{ L}} = 0.13\ M$$

$$[CO]_{eq} = \frac{0.26 \text{mol}}{0.32 \text{ L}} = 0.81\ M$$

$$[H_2]_{eq} = \frac{0.091 \text{ mol}}{0.32 \text{ L}} = 0.28\ M$$

$$[H_2O]_{eq} = \frac{[CO]_{eq}[H_2]_{eq}^3}{[CH_4]_{eq}\ K_c}$$

$$= \frac{(0.81)(0.28]^3}{(0.13)(0.26]} = 0.53\ M$$

Calculating Equilibrium Constant

Calculating K_c from concentration data

PROBLEM: In a study of hydrogen halide decomposition, a researcher fills an evacuated 2.00 L flask with 0.200 mol of HI gas and allows the reaction to proceed at 453 ºC.

$$2HI(g) \rightleftharpoons H_2(g) + I_2(g)$$

At equilibrium, $[HI]_{eq}$ = 0.078 *M*. Calculate K_c.

PLAN: Find the molar concentration of the starting material and then find the amount of reactants and products at equilibrium.

SOLUTION:

$$[HI] = \frac{0.200 \text{ mol}}{2.00 \text{ L}} = 0.100\ M$$

Let x be the amount of $[H_2]$ at equilibrium. Then x will also be the concentration of $[I_2]$ and the amount of [HI] is 2x or 0.078 *M*.

We can summarize these data in a reaction table.

A Reaction Table

concentration (*M*)	$2HI(g)$	$\rightleftharpoons$	$H_2(g)$	+	$I_2(g)$
initial	0.100		0		0
change	$-2x$		x		x
equilibrium	$0.100 - 2x$		x		x

$[HI]_{eq} = 0.078\ M = 0.100 - 2x$; $x = 0.011\ M$

$$K_c = \frac{[H_2]\,[I_2]}{[HI]^2} = \frac{[0.011]\ [0.011]}{[0.078]^2}$$

$K_c = 0.020$

Determining equilibrium concentrations from initial concentrations and K_c

PROBLEM: Fuel engineers use the extent of the change from CO and H_2O to CO_2 and H_2 to regulate the proportions of synthetic fuel mixtures. If 0.250 mol of CO and 0.250 mol of H_2O are placed in a 125 mL flask at 900 K, what is the composition of the equilibrium mixture? At this temperature, K_c is 1.56 for the equation,

$$CO(g) + H_2O(g) \rightleftharpoons CO_2(g) + H_2(g)$$

PLAN: Find the concentrations of all species at equilibrium and then substitute into a K_c expression.

SOLUTION: Initial concentrations must be calculated as *M*,

so [CO] = [H_2O] = **0.250/0.125L**.=2.M

concentration	$CO(g)$	$+ H_2O(g)$	$\rightleftharpoons CO_2(g)$	$+ H_2(g)$
initial	2.00	2.00	0	0
change	$-x$	$-x$	x	x
equilibrium	$2.00 - x$	$2.00 - x$	x	x

$$K_c = \frac{[CO_2][H_2]}{[CO][H_2O]} = \frac{(x)\quad(x)}{(2.00 - x)\,(2.00 - x)} = \frac{(x)^2}{(2.00 - x)^2} = 1.56$$

$$\sqrt{1.56} = \frac{x}{2.00 - x}$$

$$1.25 = \frac{x}{2.00 - x}$$

$x = 1.11\ M$

$[CO_2] = [H_2] = 1.11\ M$

$2.00 - x = 0.89\ M$

$[CO] = [H_2O] = 0.89\ M$

Predicting reaction direction and calculating equilibrium concentrations

PROBLEM: The research and development unit of a chemical company is studying the reaction of CH_4 and H_2S, two components of natural gas.

$$CH_4(g) + 2H_2S(g) \rightleftharpoons CS_2(g) + 4H_2(g)$$

In one experiment, 1.00 mol of CH_4, 1.00 mol of CS_2, 2.00 mol of H_2S and 2.00 mol of H_2 are mixed in a 250 mL vessel at 960 ºC. At this temperature, K_c = 0.036.

(a) In <u>which direction</u> will the reaction proceed to reach equilibrium?

(b) If $[CH_4]$ = 5.56 *M* <u>at equilibrium</u>, what are the equilibrium concentrations of the other three substances?

PLAN: Find the initial molar concentrations of all components and use these to calculate K_c.

SOLUTION:

$[CH_4]_{initial}$ = 1.00 mol/0.25 L = 4.00 *M*

$[CS_2]_{initial}$ = 1.00 mol/0.25 L = 4.00 *M*

$[H_2S]_{initial}$ = 2.00 mol/0.25 L = 8.00 *M*

$[H_2]_{initial}$ = 2.00 mol/0.25 L = 8.00 *M*

K « 1 ⇒ small *K*

K_c = **0.036**

The reaction will progress to the left.

left

concentrations	$CH_4(g)$ +	$2H_2S(g)$ ⇌	$CS_2(g)$ +	$4H_2(g)$
initial	4.00	8.00	4.00	8.00
change	+ *x*	+ 2*x*	- *x*	- 4*x*
equilibrium	4.00 + *x*	8.00 + 2*x*	4.00 - *x*	8.00 - 4*x*

At equilibrium $[CH_4]$ = 5.56 *M*, so 5.56 = 4.00 + *x*; thus, *x* = 1.56 *M*

Therefore:

$[H_2S]$ = 8.00 + 2*x* = 11.12 *M*

$[H_2]$ = 8.00 - 4*x* = 1.76 *M*

$[CS_2]$ = 4.00 - *x* = 2.44 *M*

Calculating Equilibrium Constant

- Example-1:

  0.2 mole of H_2 gas and **0.2 mole** of I_2 vapor are introduced into a **one-liter** sealed flask. The mixture is heated to a certain temperature and the following reaction occurs until equilibrium is established.

$$H_2(g) + I_2(g) \rightleftarrows 2HI(g)$$

  At equilibrium, the mixture is found to contain **0.316 mole** of HI.

(a)What are the concentrations of H_2, I_2 and HI at equilibrium?

(b)Calculate the equilibrium constant K_c.

Calculating Equilibrium Constant for reaction: $H_{2(g)} + I_{2(g)} \rightleftarrows 2HI_{(g)}$

	$H_2(g)$	+	$I_2(g)$	$\rightleftarrows$	2 HI(g)
Initial [ ],	0.200		0.200		0.000
Change in [ ],	-x		-x		+2x
Equilibrium[ ],	0.200 - x		0.200 - x		+2x

- **At equilibrium** [HI] is **0.316 mole**
- *2x = 0-316 X = 0. 158 M*

- $[H_2]$, = $[I_2]$ = 0.200-x
- = 0.200 - *0.158* = 0.042M

- $$K_c = \frac{[HI]^2}{[H_2][I_2]} = \frac{(0.316)^2}{(0.042)^2} = 0.09986 / 0.001764$$

$$= 56.6$$

Equilibria Involving Weak Acids and Bases

Consider acetic acid, $HC_2H_3O_2$ (HOAc)

$$HC_2H_3O_2 + H_2O \rightleftharpoons H_3O^+ + C_2H_3O_2^-$$

Acid Conj. base

$$HA(aq) + H_2O(\ell) \rightleftharpoons H_3O^+(aq) + A^-(aq)$$

$$K_a = \frac{[H_3O^+][A^-]}{[HA]}$$

K is designated K_a for ACID)

K gives the ratio of ions to molecules

Weak acid has $K_a < 1$

Leads to small $[H_3O^+]$ and a pH of 2 - 7

Equilibrium Constants for Weak Bases

$$B(aq) + H_2O(\ell) \rightleftharpoons BH^+(aq) + OH^-(aq)$$

$$K_b = \frac{[BH^+][OH^-]}{[B]}$$

Weak base has $K_b < 1$

Leads to small $[OH^-]$ and a pH of 12 - 7

Equilibria Involving A Weak Acid

You have 1.00 M HOAc. Calc. the equilibrium concs. of HOAc, H_3O^+, OAc^-, and the pH. Ka= 1.8 x 10-5

Step 1. Define equilibrium concs. in ICE (initial Change equilib)table.

	[HOAc]	[H_3O^+]	[OAc^-]
initial	1.00	0	0
change	-x	+x	+x
Equilibrium	1.00-x	x	x

Equilibria Involving A Weak Acid

Step 2. Write K_a expression

$$K_a = 1.8 \times 10^{-5} = \frac{[H_3O^+][OAc^-]}{[HOAc]} = \frac{x^2}{1.00 - x}$$

you can make an approximation if x is very small!

Equilibria Involving A Weak Acid

Step 3. Solve K_a expression

$$K_a = 1.8 \times 10^{-5} = \frac{[H_3O^+][OAc^-]}{[HOAc]} = \frac{x^2}{1.00 - x}$$

First assume x is very small because K_a is so small.

$$K_a = 1.8 \times 10^{-5} = \frac{x^2}{1.00}$$

Now we can more easily solve this approximate expression.

Equilibria Involving A Weak Acid

Step 3. Solve K_a approximate expression

$$K_a = 1.8 \times 10^{-5} = \frac{x^2}{1.00}$$

$$x = [H_3O^+] = [OAc^-] = 4.2 \times 10^{-3}\ M$$

$$pH = -\log [H_3O^+] = -\log (4.2 \times 10^{-3}) = 2.37$$

Equilibria Involving A Weak Base

You have 0.010 M NH_3. Calc. the pH.

$$NH_3 + H_2O \rightarrow NH_4^+ + OH^-$$

$K_b = 1.8 \times 10^{-5}$

Step 1. Define equilibrium concs. in ICE table

	$[NH_3]$	$[NH_4^+]$	$[OH^-]$
initial	0.010	0	0
change	-x	+x	+x
equilib	0.010 - x	x	x

Equilibria Involving A Weak Base

$K_b = 1.8 \times 10^{-5}$

Step 2. Solve the equilibrium expression

$$K_b = 1.8 \times 10^{-5} = \frac{[NH_4^+][OH^-]}{[NH_3]} = \frac{x^2}{0.010 - x}$$

Assume x is small, so

$$x = [OH^-] = [NH_4^+] = 4.2 \times 10^{-4}\ M$$

and $[NH_3] = 0.010 - 4.2 \times 10^{-4} \approx 0.010\ M$

Equilibria Involving A Weak Base

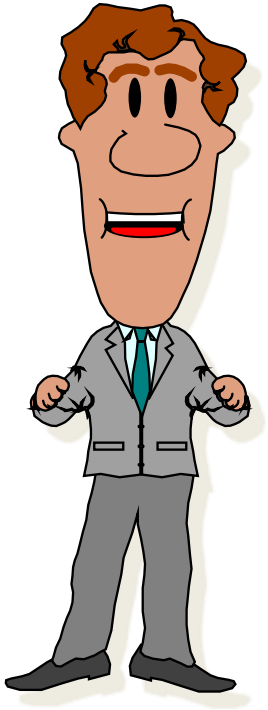
Step 3. Calculate pH

$[OH^-] = 4.2 \times 10^{-4}$ M

so pOH = - log $[OH^-]$ = 3.37

Because pH + pOH = 14,

pH = 10.63



Calculating pka:

The pka calculating using this expression

pka= -log[ka]

Where (ka) is equilibrium constant for the acid.

Example:

What is the pka of acetic acid if ka for acetic acid is $1.78X10^{-5}$?

pka = $-\log(1.78X10^{-5})$=

$-(-4.75)$=4.5?

Calculating ka from the pka:

The ka for and acid is calculating from the pka by performing the reverse of the mathematical operation used to find pka

$Ka=10^{-pka}$ or ka=antilog(-pka)

Example:

Calculating the value of the ionization constant for the ammonium ion, ka if the pka is 9.74?

$9.74 = -\log(ka)$

$-9.74 = \log(ka)$

$ka = 10^{-9.74} = 1.82 \times 10^{-10}$

On to calculator calculate

$10^{-9.74}$ or "in verse"

$\log(-9.74)$.